Detecting Sub-Millimetre Bone Fractures
via the X-ray Scatter-to-Primary Ratio

A Monte Carlo (OpenGATE 10 / Geant4) Feasibility Study

[Author Name] • Baanwit Academy • 2026 • *preliminary — CL-3 & CL-6 pending*

1. Introduction

- Hairline / sub-millimetre fractures are frequently **invisible on plain radiographs**: the attenuation contrast they create is tiny.
- **Hypothesis:** a fracture perturbs the local scatter field, leaving a signature in the **Scatter-to-Primary Ratio**

$$\mathrm{SPR}(x,y) = \frac{I_{\text{scatter}}}{I_{\text{primary}}}$$

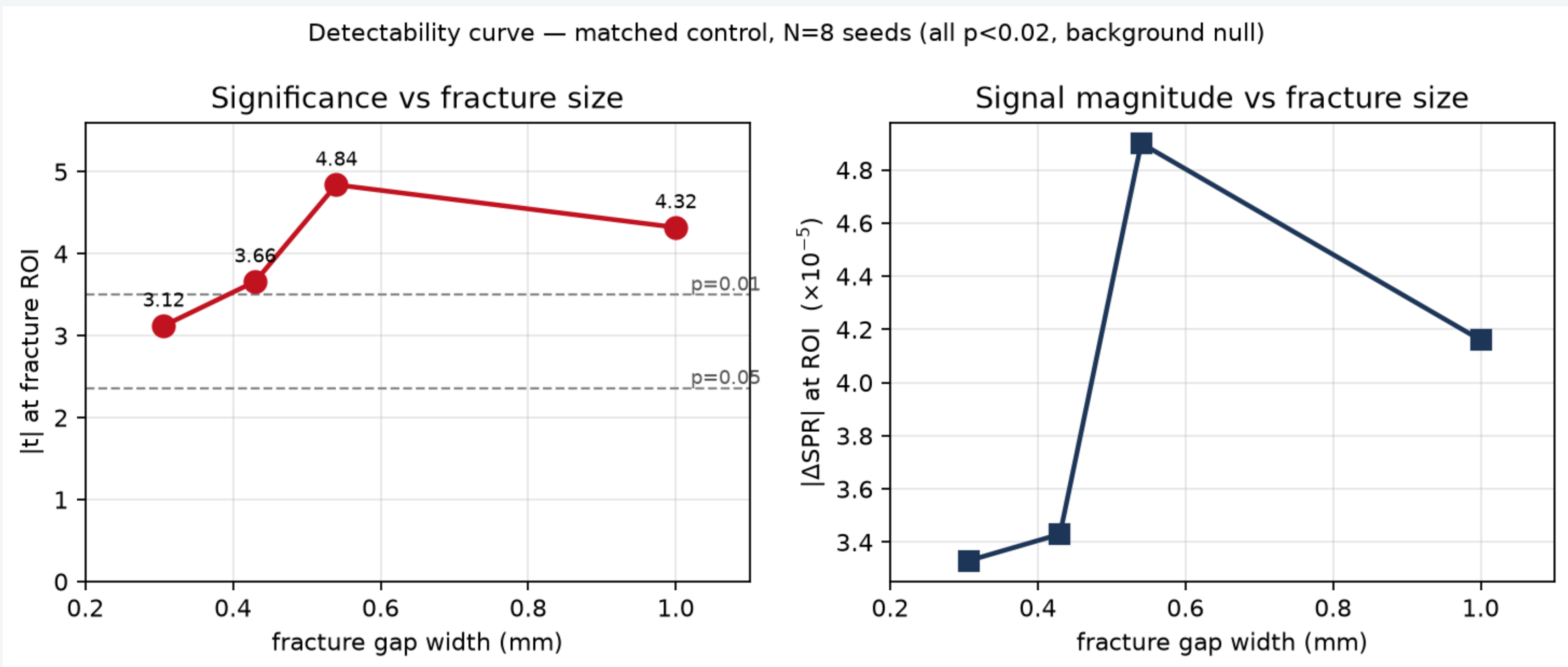
- **Aim:** test by Monte Carlo whether this SPR signature is *detectable*, *statistically robust*, *not an artifact*, and *physically grounded.*

2. Methods

- **Engine:** OpenGATE 10 (Geant4), Livermore physics; 80 keV; compact bone (ICRU); AP pelvis; SPR map from primary/scatter separation at the detector.
- **Fracture models:** gap widths 0.306, 0.43, 0.54, 1.0 mm.
- **Matched control (key):** the *same* mesh with the gap filled ⇒ $\Delta\mathrm{SPR} = \mathrm{SPR}_{\text{frac}} - \mathrm{SPR}_{\text{ctrl}}$ isolates **only the fracture** (removes the mesh-tessellation confound).
- **Multi-seed significance:** N=8 independent seeds, common random numbers; *a-priori* ROI t-test at the fracture vs. a background ROI.
- **Collimation** (40 mm field) concentrates photon statistics on the fracture.

3. Detection (CL-0)

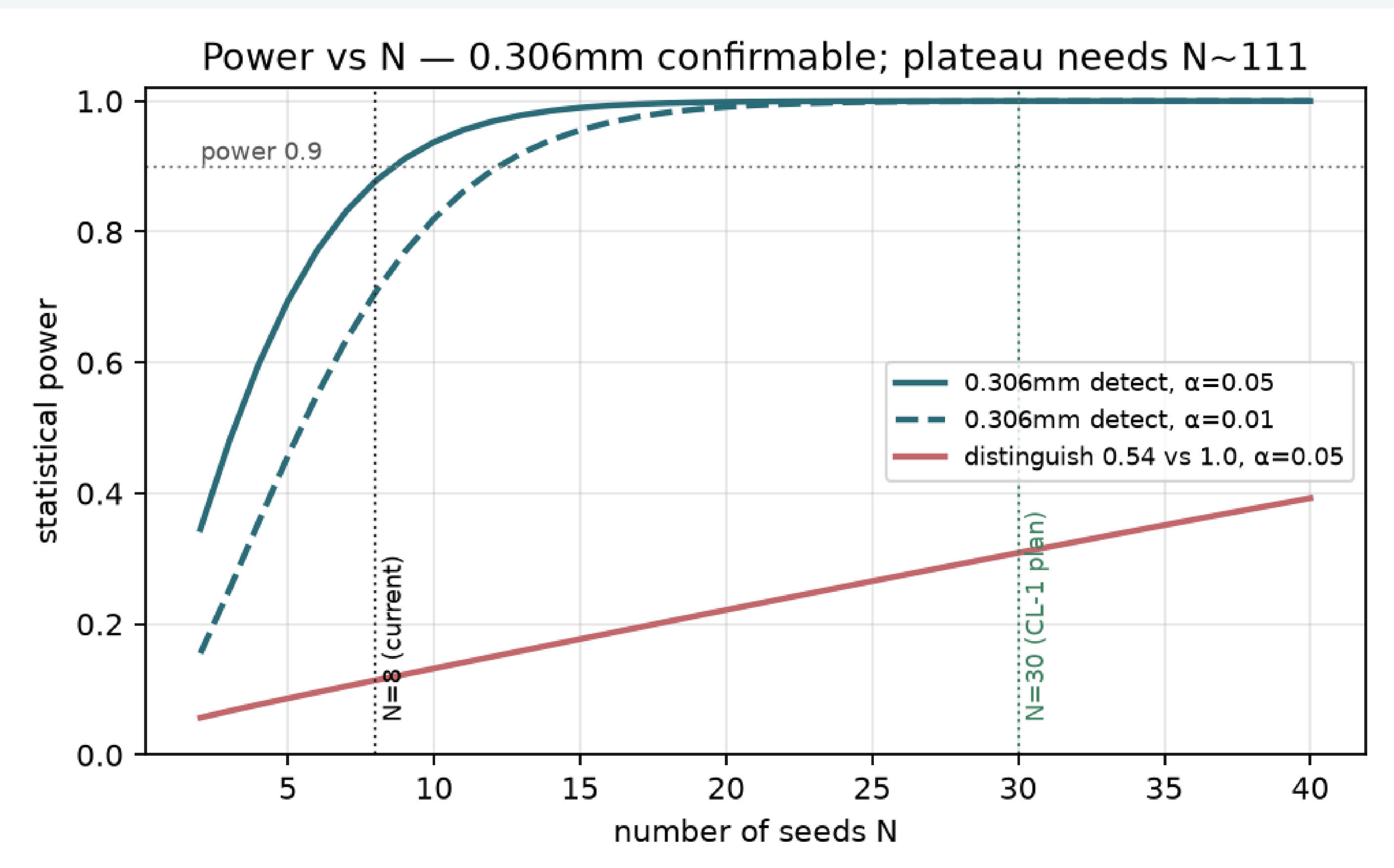
All four sizes detected ($p < 0.02$); $\Delta\mathrm{SPR} < 0$ (air gap → more primary → lower SPR); background ROI **null** ⇒ the signal is localised to the fracture.



Signal rises with gap width from 0.306 to 0.54 mm; $|t| = 3.1, 3.7, 4.8, 4.3$.

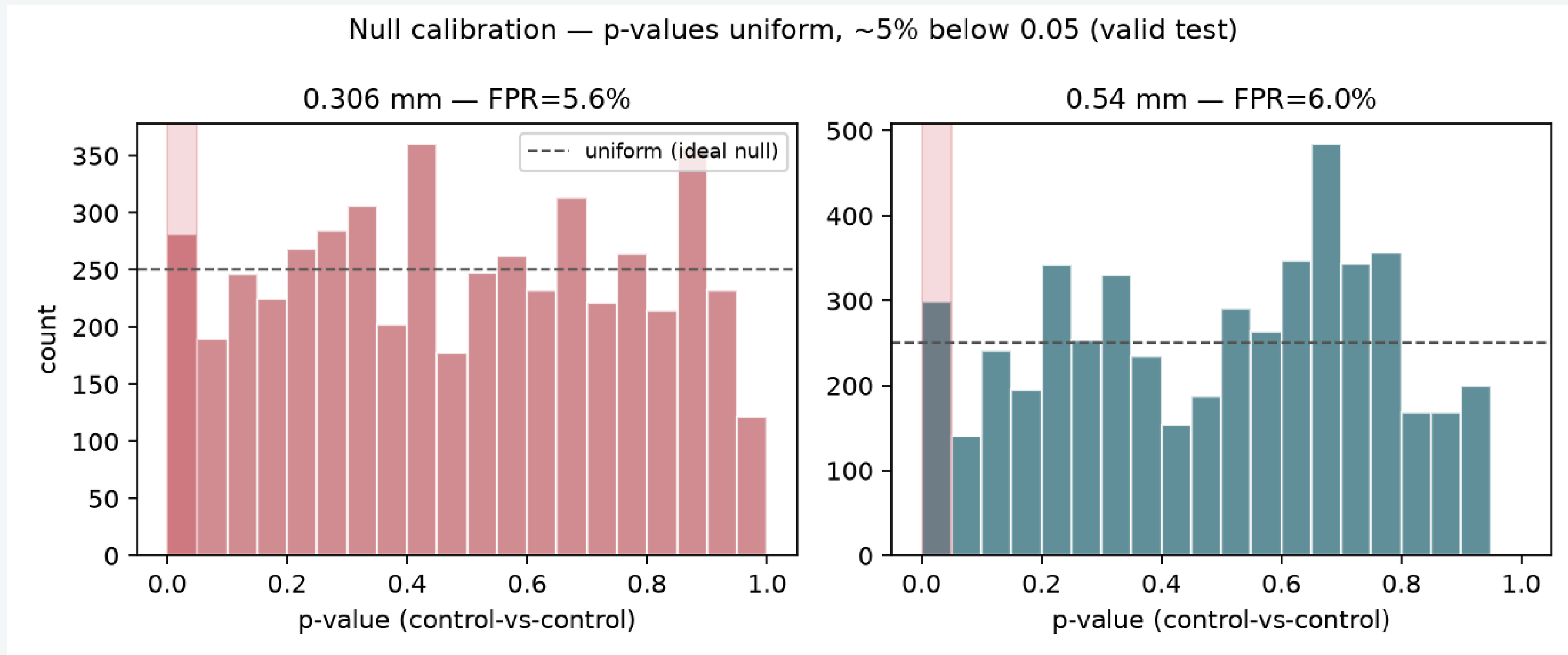
4. Robustness & Power (CL-1)

- Bootstrap 95% CIs **exclude 0** for all sizes ⇒ detections are real.
- Smallest fracture (0.306 mm) is **borderline** (80% reliability at N=8); ∼13 seeds reach $p < 0.01$.



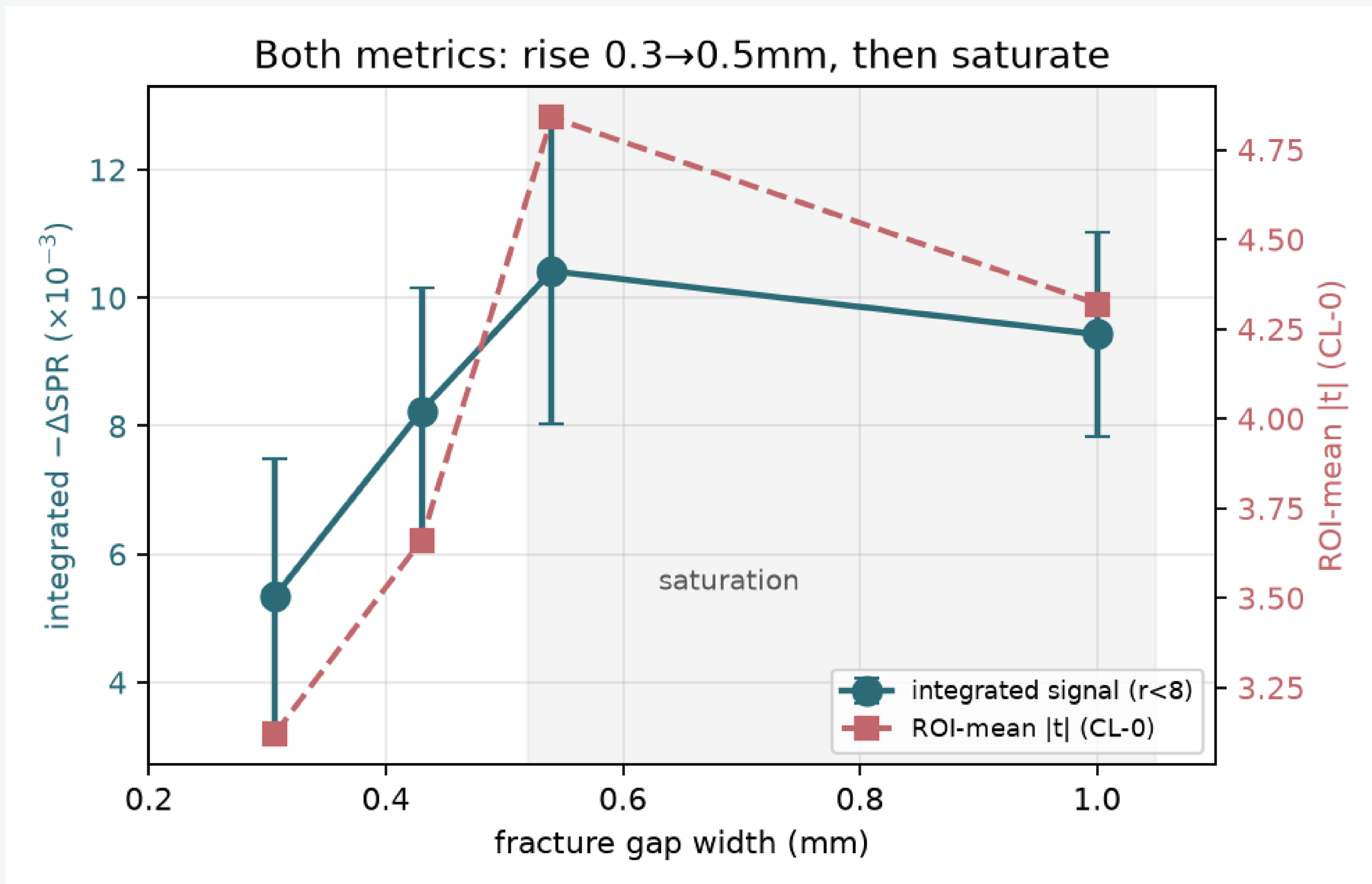
5. Validity — no false positives (CL-2)

- Control-vs-control false-positive rate = 5.6–5.8% ≈ nominal 5%.
- Sign-flip permutation confirms significance non-parametrically.
- Only ∼5% of in-beam pixels significant (chance) ⇒ **no global artifact**.



6. Plateau resolution (CL-7)

- Signal rises 0.306→0.54 mm then **saturates** (0.54 ≈ 1.0, p=0.73).
- **Integrated** (spatial) signal saturates too ⇒ saturation is **physical**, not an ROI artifact.



7. Mechanism tests — in progress

CL-3 Energy dependence (40–120 keV): does the signal follow photoelectric/Compton physics? **[running]**

CL-6 Beam-angle dependence: is the signal maximal when the beam is tangential to the fracture plane (the clinical hallmark)? **[running]**

Results to be inserted in the final version; per-angle geometry pre-computed and queued.

8. Conclusions

- The SPR carries a **real, validated** fracture signature detectable to ∼0.3 mm.
- The signal is **physically bounded**: excellent for *detection*, saturating for *width* > 0.5 mm.
- Energy- & angle-dependence (pending) will complete the mechanistic proof for **Phase 1**.

9. Phase-1 evidence pillars

Detection	Robustness	Validity	Phys. bound	Energy	Geometry
CL-0	CL-1	CL-2	CL-7	CL-3	CL-6
✓	✓	✓	✓	running	running